Digital Methods for Calculating the Area of Sunspots

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Introduction

Sunspots are darker regions on the Sun's surface linked to magnetic field activity. Accurately calculating their areas is essential for understanding solar activity and its potential impacts on space weather. This poster examines methods for calculating sunspot areas through image processing and corrections for heliocentric angles. Images were captured with a Celestron NexStar 8SE telescope and a Nikon DSLR D650, with additional comparisons made using images from the Solar Influences Data Analysis Center (SIDC).

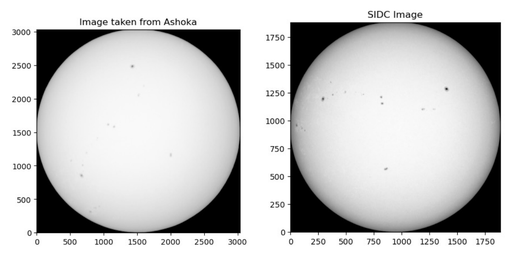


Fig. 1: Images for comparison

Adaptive Threshold Method to make it black and white!

A black-and-white image could help us see the darker spots. Everything dark is 0 and bright is 1.

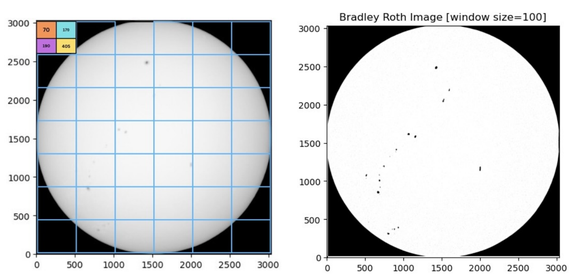


Fig. 2: Break Down Image into squares. Take the average 'pixel count' and assign each pixel 1/0

This resulted in a Trial method where the size of the spot depended heavily on window size and threshold value. **Can we do it without Brute forcing to find the ideal parameters?**

Limb Darkening

The Sun's intensity profile peaks at the center and tends to zero as we cross its edge. It messes with thresholding as the sun is not 'uniformly bright'

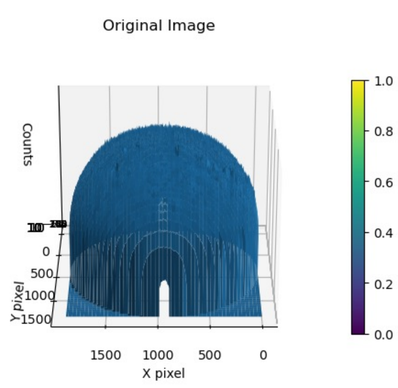


Fig. 3: 3D plot of the image and it's pixel counts (value assigned based on amount of light hitting the pixel)

1. First draw circles on the sun at different radii- creating a radial intensity profile
2. Fit the radial intensity to a polynomial
3. Divide each pixel with the corresponding polynomial value

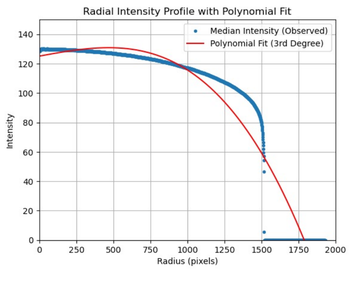


Fig. 4: There is sharp drop towards the end as the images were cropped such that the border is exactly at the edge of the sun

Limb Darkening Continued

Doing so results in a normalized image, much smoother than the earlier results.

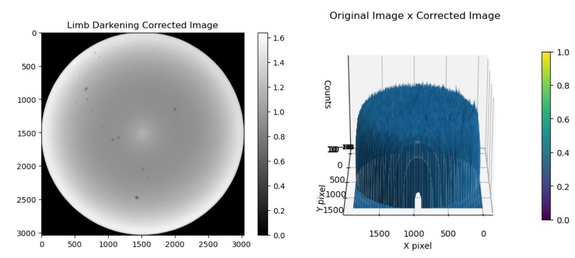


Fig. 5: The Normalized Image alongside a 3D plot of the images pixel counts

Thresholding Normalized Image

Example of roughly thresholded image (optimum window size found to be 50 or 100)

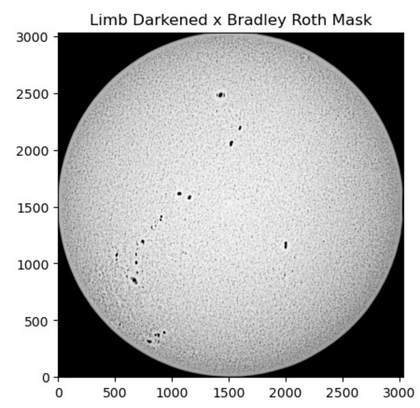


Fig. 6: It looks pretty grainy, and that is noise. We will take care of that too in a minute

Morphological Processing

We use two techniques in morphological processing to reduce noise while maintaining key features: **opening** and **closing**. Opening smooths edges by shrinking objects (erosion) and then expanding them (dilation), effectively removing small spots. Conversely, closing fills gaps by expanding objects and then shrinking them back down, resulting in cleaner, more defined images.

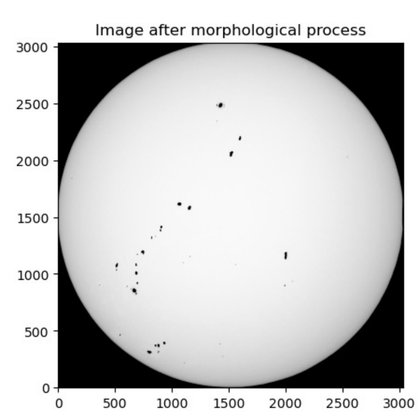
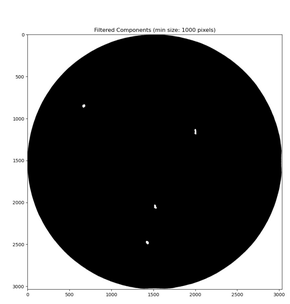


Fig. 7: We can see that the noise has reduced, and the spots themselves are clearer

Connected Components - Measuring the Area

Measured area using connected components. Looking for large areas of value = 0



Connected Components

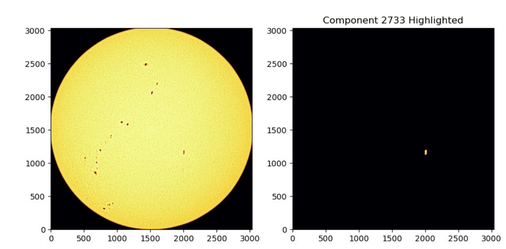


Fig. 9: Here we see all spots with area 1000 pixels or greater

Correcting for Heliogroaphic angle

find the heliocentric angle ρ of a sunspot calculate the angle between the line of sight and the sunspot's position.

$$\rho = \cos\left(\frac{r}{R_\odot}\right)$$

where

- $r = \sqrt{x^2 + y^2}$ is the distance of the sunspot from the disk center in pixels,
- $R_\odot$ is the solar disk radius in the image, in pixels.

Corrected Sunspot Area

Adjust the observed area A_{obs} in pixels for foreshortening based on the heliocentric angle.

$$A_{\text{corrected}} = \frac{A_{\text{obs}}}{\sqrt{1 - \left(\frac{y}{R_\odot}\right)^2 - \left(\frac{z}{R_\odot}\right)^2}}$$

y- projection along the solar equator
z is the projection perpendicular to it.

Conversion to Physical Units

From pixel to km^2 to MH - (Millionths of Hemisphere). 1 MH = 3 million km^2

Results

Area results compared to spaceweatherlive.com.

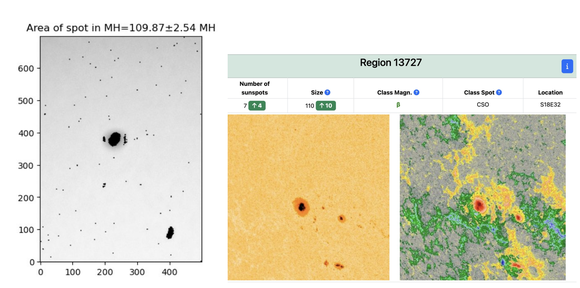


Fig. 10: Our image vs. spaceweatherlive.com(Note our image is flipped)

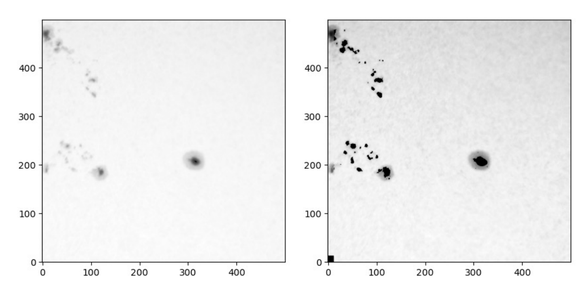


Fig. 11: Image process repeated for SIDC image

Using the SIDC images we can show we are able to isolate spot. We have almost the same resolving power as them, it's the atmosphere around Sonipat that affects the clarity

References

1. "Journal of Astrophysics and Astronomy | Indian Academy of Sciences." n.d.
2. Solar Influences Data and Analysis center
3. Bradley and Roth 2007